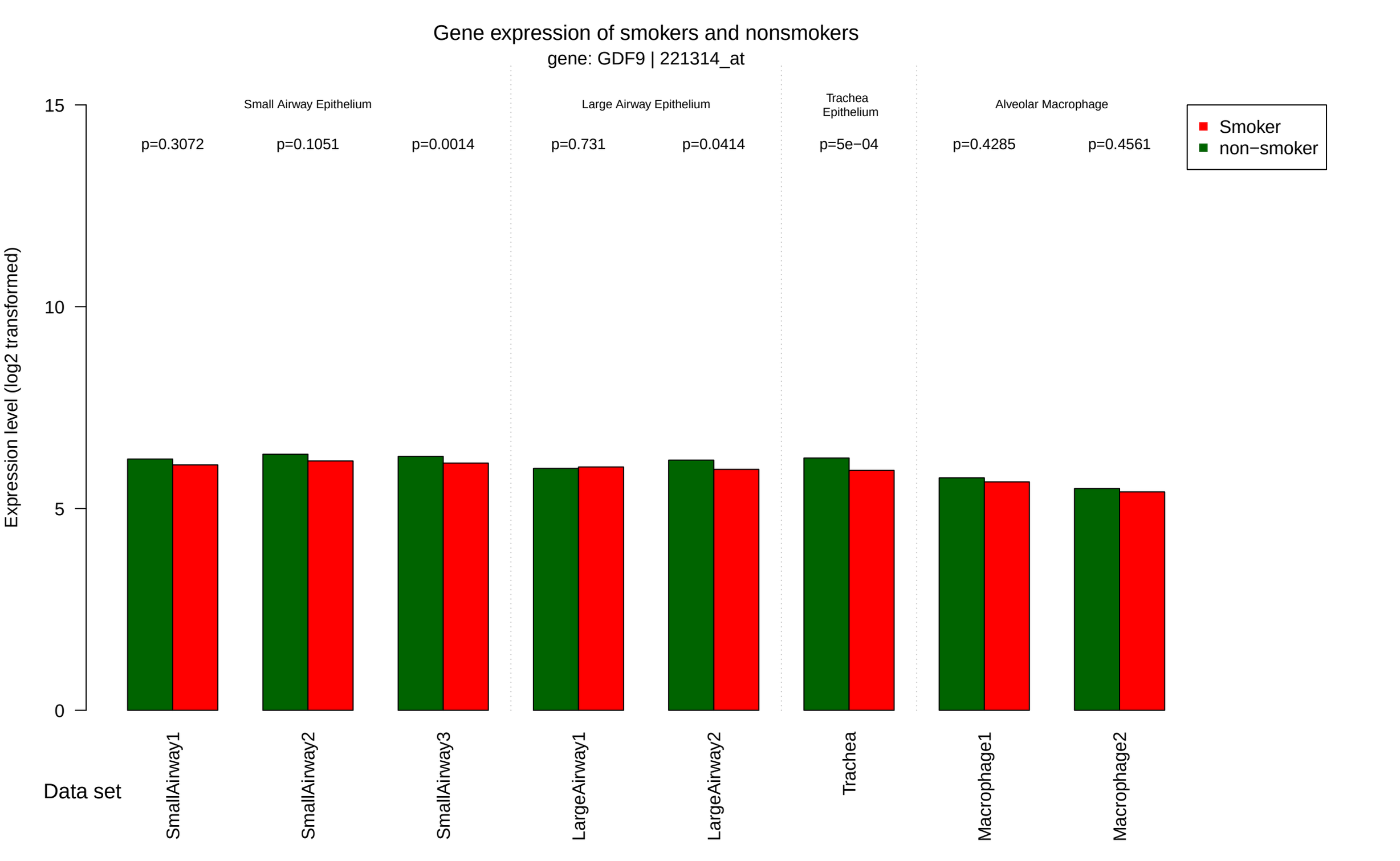
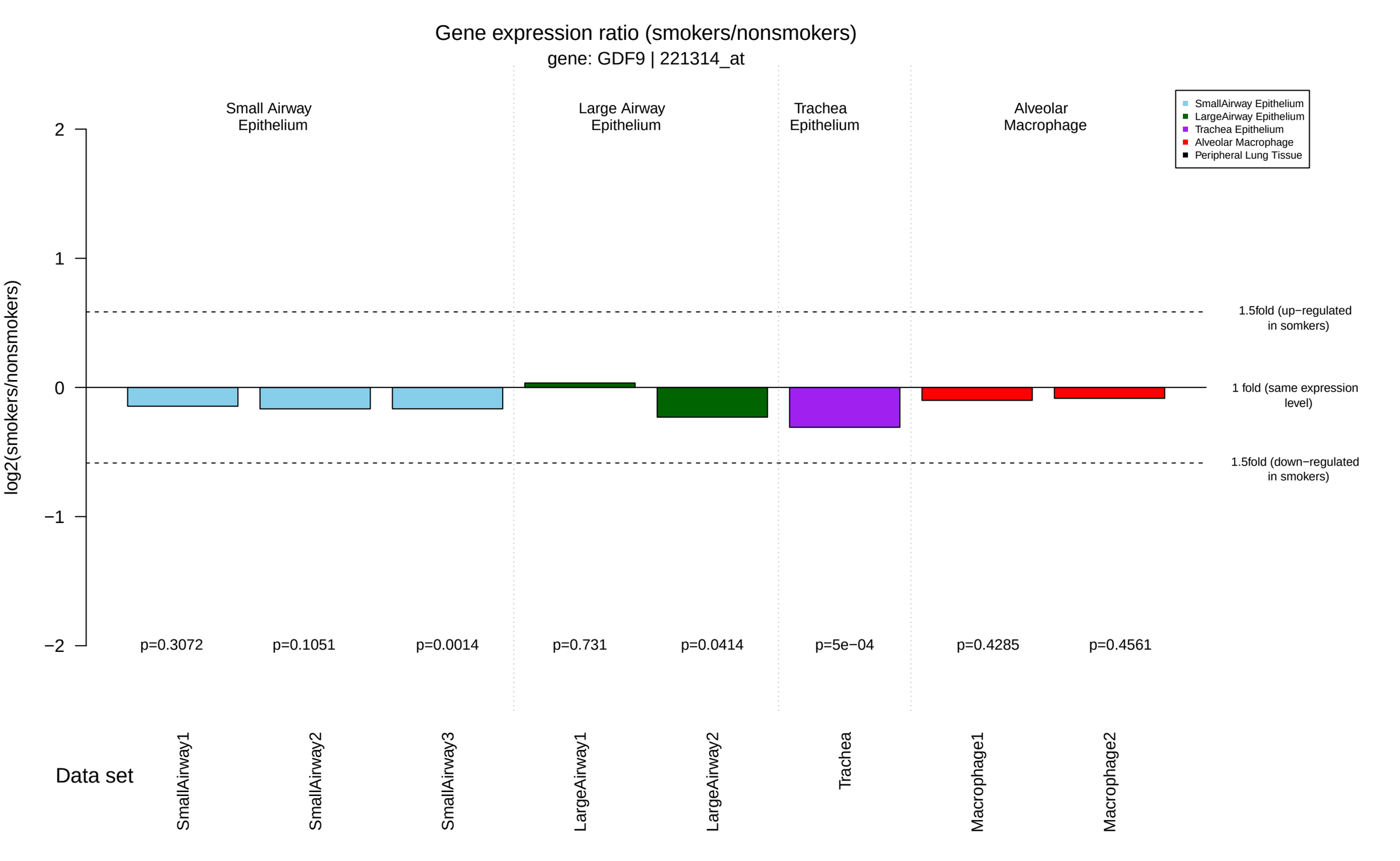
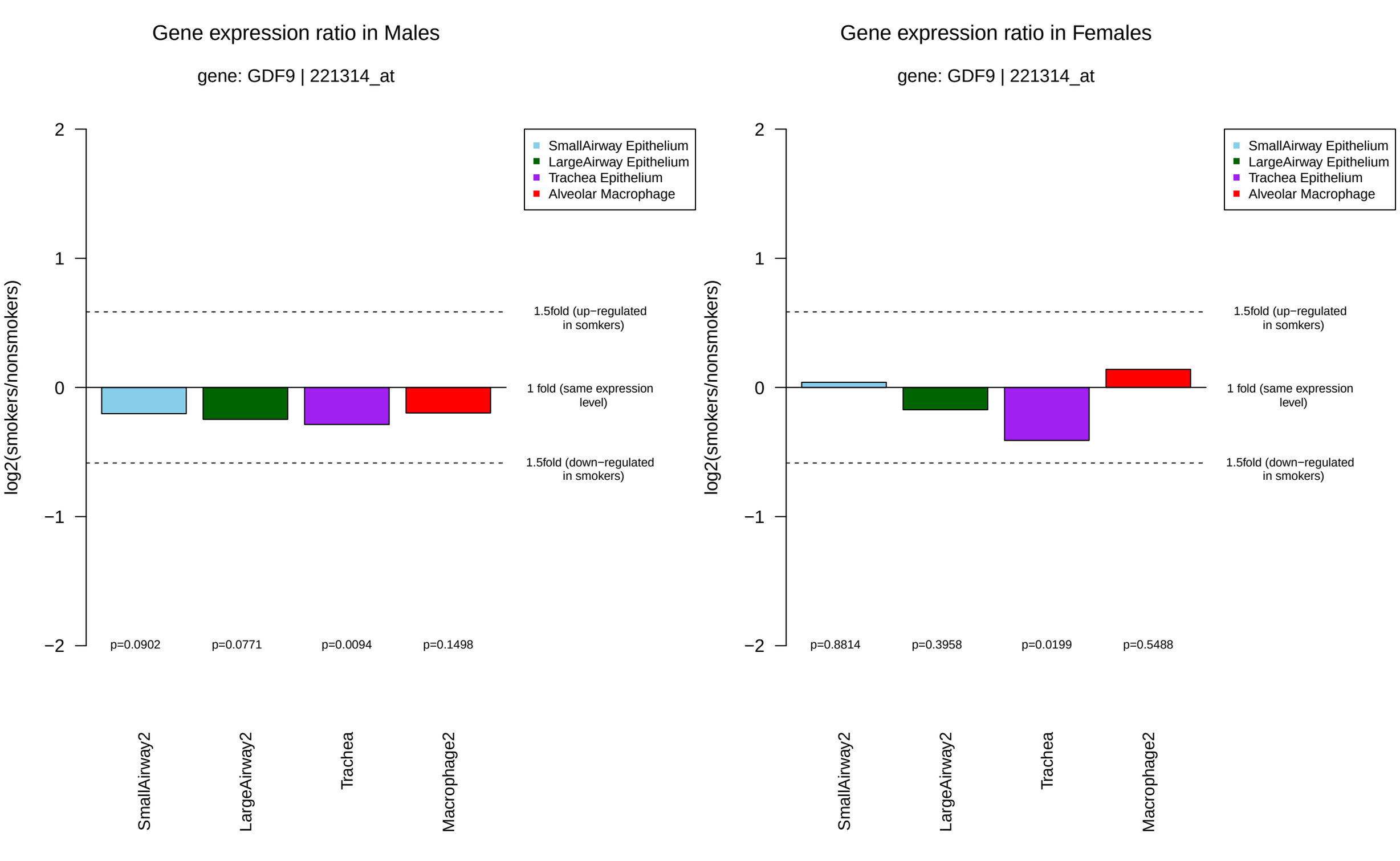
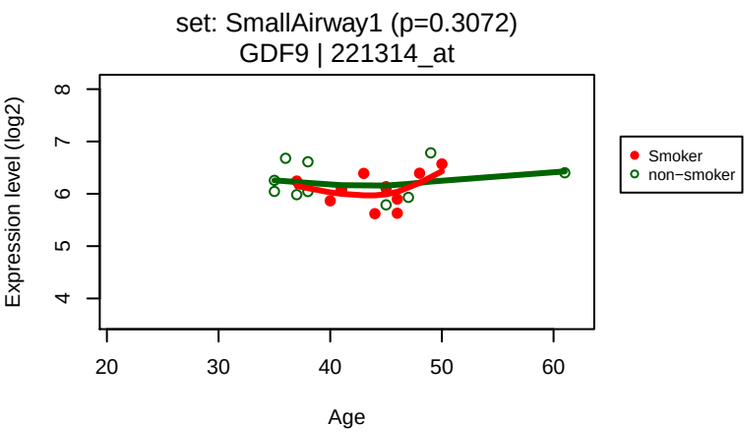
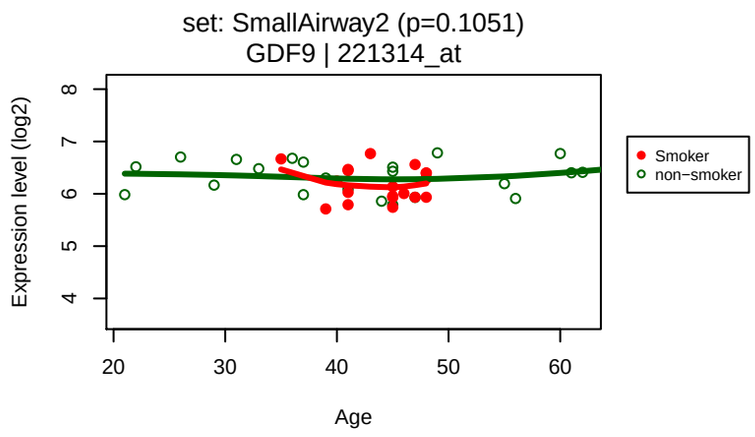
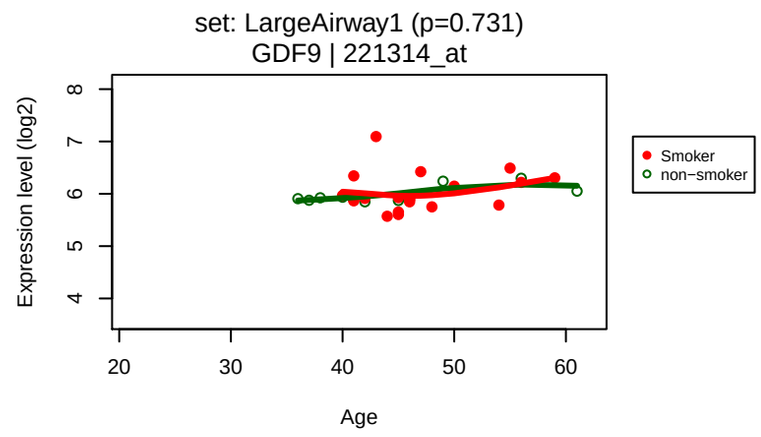

Gene expression of smokers and nonsmokers
gene: GDF9 | 221314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3072
p=0.1051
p=0.0014
p=0.731
p=0.0414
p=5e−04
p=0.4285
p=0.4561
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

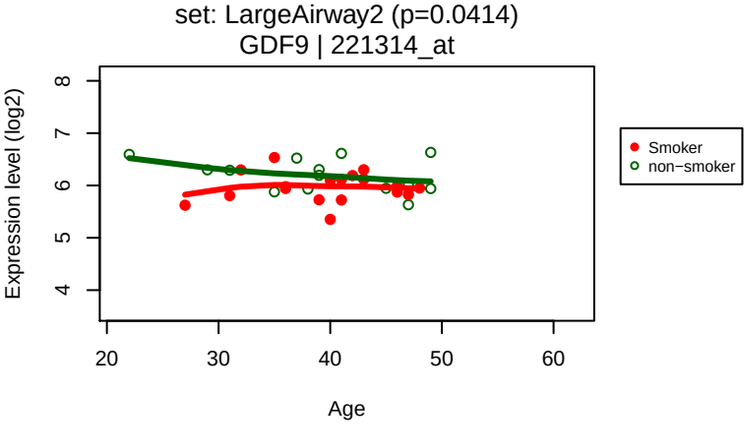

Gene expression ratio (smokers/nonsmokers)
gene: GDF9 | 221314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3072
p=0.1051
p=0.0014
p=0.731
p=0.0414
p=5e−04
p=0.4285
p=0.4561
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

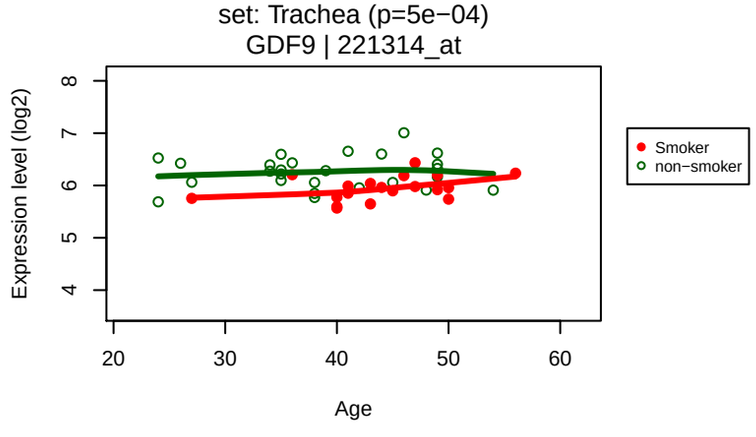

Gene expression ratio in Males
gene: GDF9 | 221314_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0902
p=0.0771
p=0.0094
p=0.1498
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GDF9 | 221314_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8814
p=0.3958
p=0.0199
p=0.5488
SmallAirway2
LargeAirway2
Trachea
Macrophage2

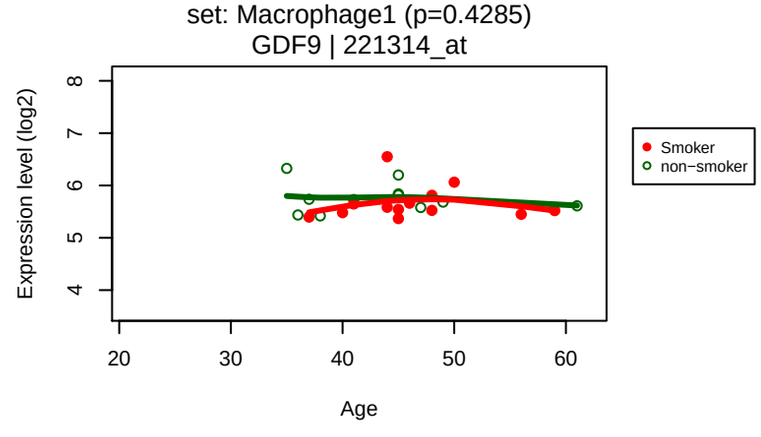

set: SmallAirway1 (p=0.3072)
GDF9 | 221314_at
Expression level (log2)
Age
Smoker
non−smoker

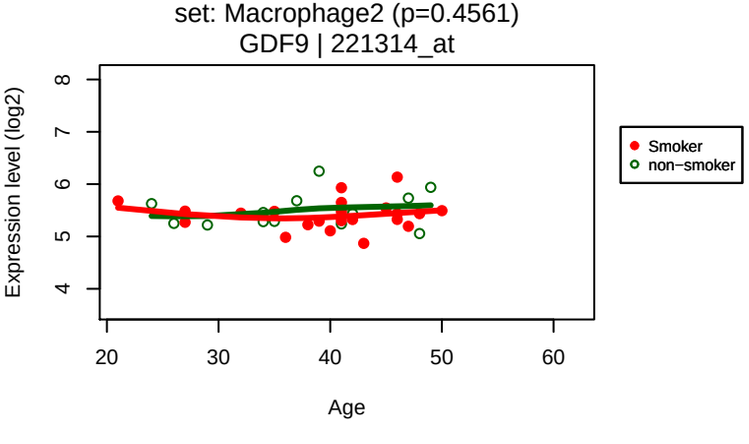

set: SmallAirway2 (p=0.1051)
GDF9 | 221314_at
Expression level (log2)
Age
Smoker
non−smoker

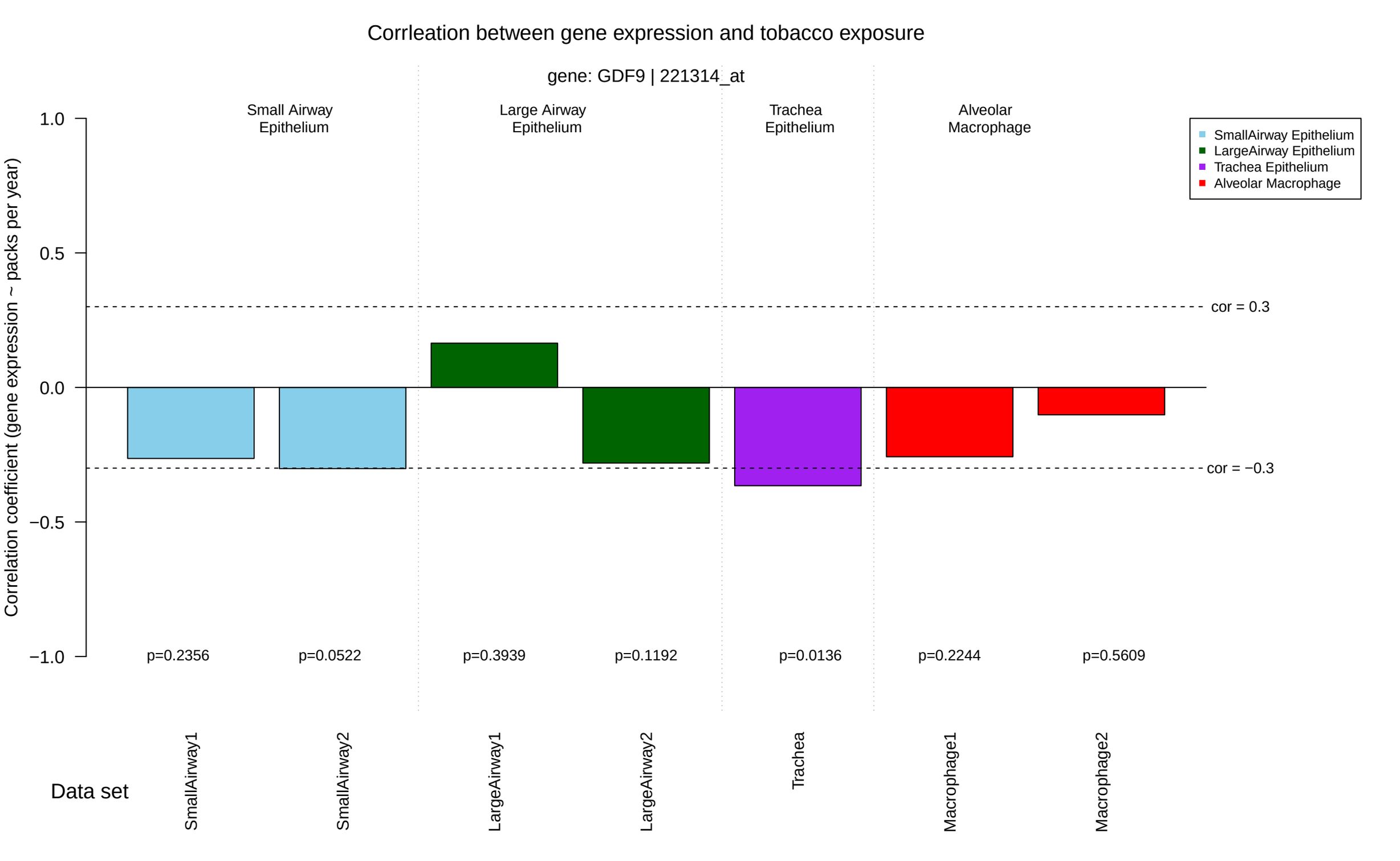

set: LargeAirway1 (p=0.731)
GDF9 | 221314_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0414)
GDF9 | 221314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
GDF9 | 221314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4285)
GDF9 | 221314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4561)
GDF9 | 221314_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GDF9 | 221314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2356
p=0.0522
p=0.3939
p=0.1192
p=0.0136
p=0.2244
p=0.5609
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2